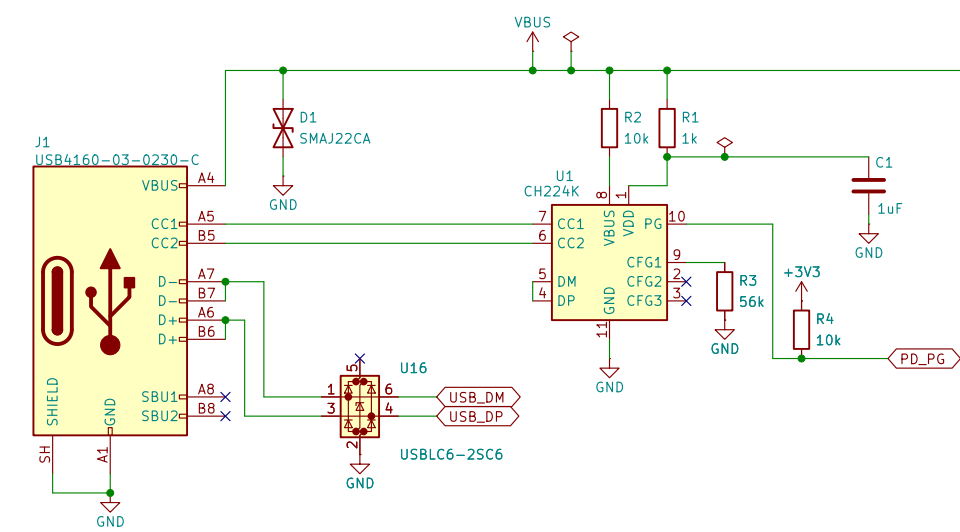
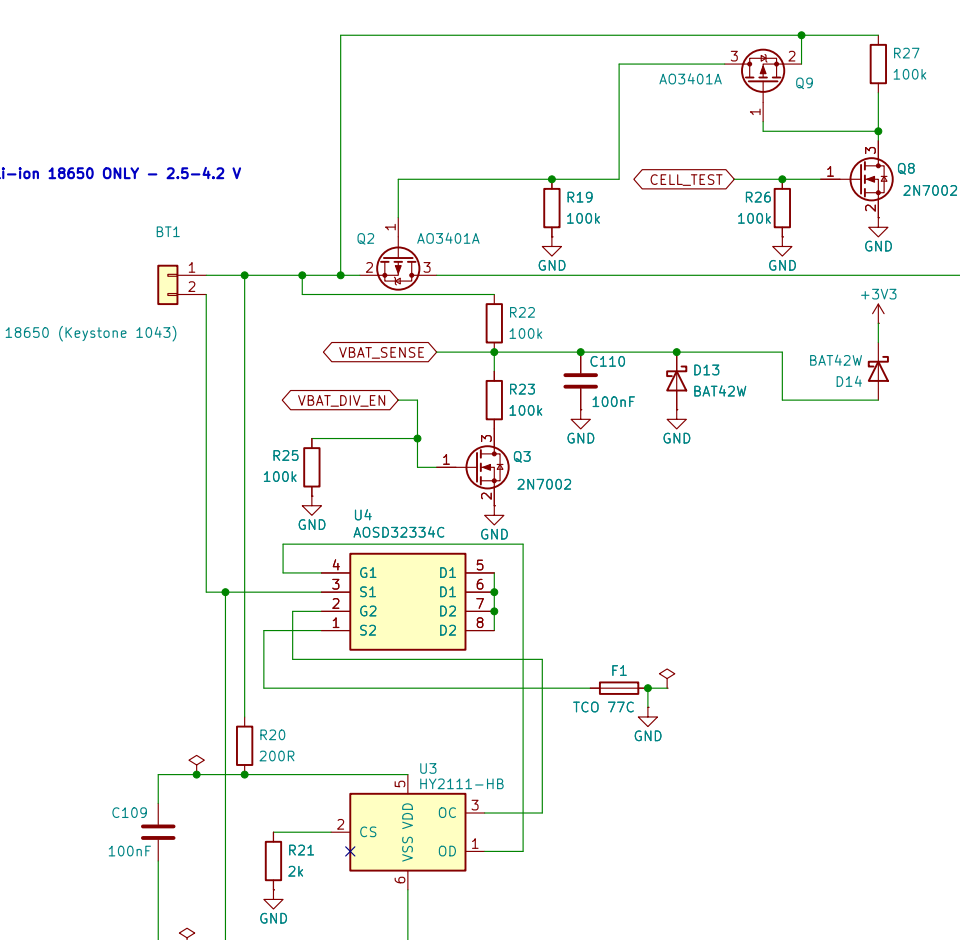


POWER IN – USB-C PD sink (15 V)



CH224K DP/DM shorted = PD-only: Type-C Ds go to the MCU.
CFG4=564 -> 15 V PDD; falls back to 5 V (charger idles <11.2 V).
USB 2.0 subset of the 24-pin USB4160 (SS pads unconnected).
D+/D- = native USB-JTAG (0019/20). Vertical – port out the back wall.

BATTERY – 18650 + reverse P-FET + protector + TCO

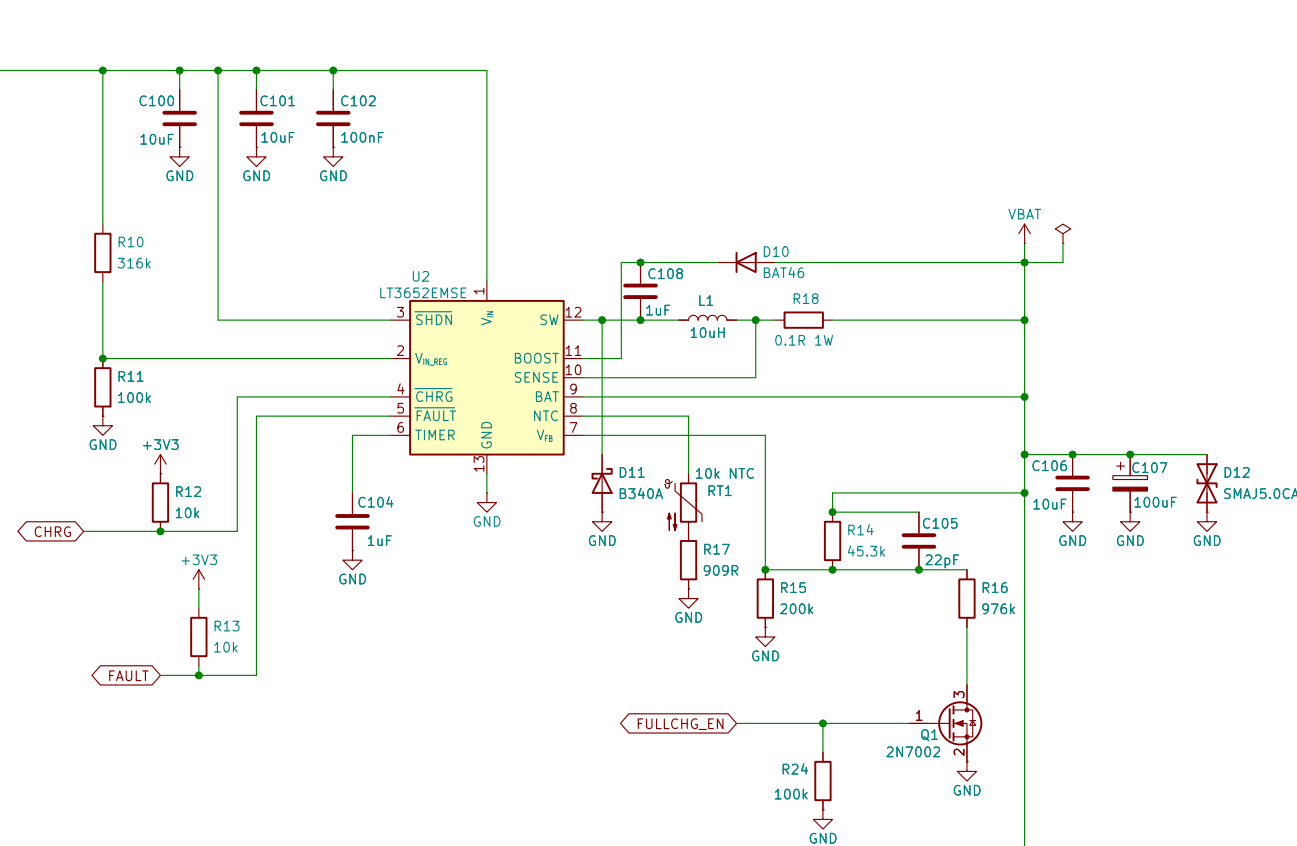


cell- -> A0503233AC (00/0C gated) -> TCO 77 C -> PACK- (GND)
0V 4.28 V / 00 2.90 V fixed (-G8). Independent of the charger.

HOW TO READ THIS SHEET

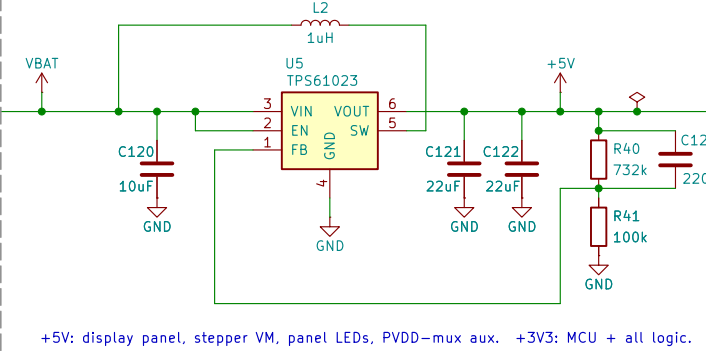
One page, laid out like block_diagram.drawio: power flows left.
MCU centre, peripherals right. Power paths and MCU buses (SPI, I2S, MCPWM, encoder) are drawn as wires; GND / +3V3 / +5V taps and named labels connect by name (same name = same net).
Off-board parts (SD card, sensors, speaker, wake-LED strips, knob, radio toggle, cell) enter via connectors J1/J11 / BT1.

CHARGER – LT3652 (15 buck, 4.05 V float, 1.0 A)



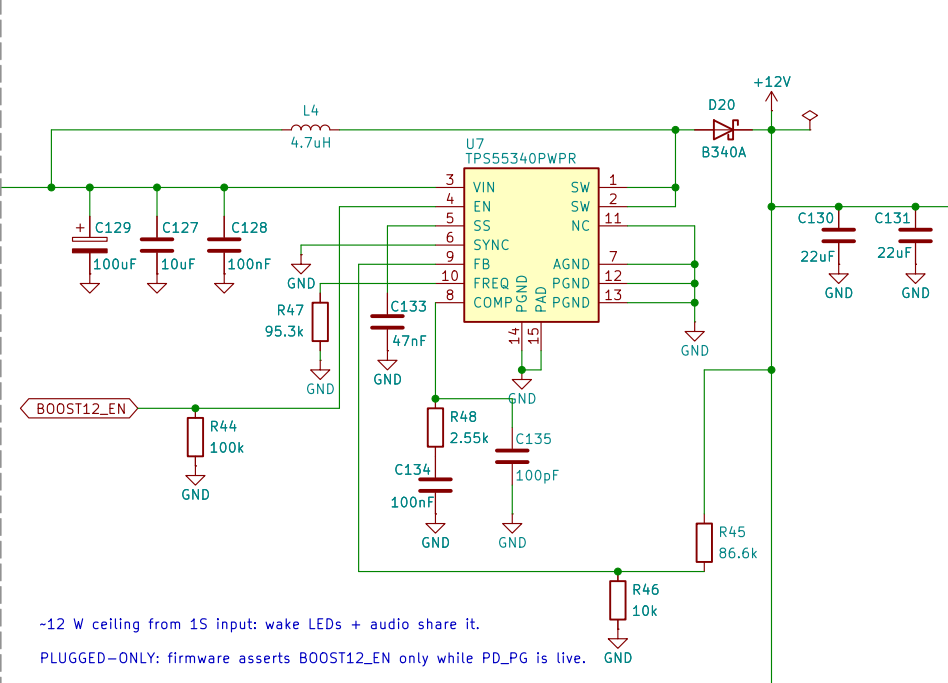
Float 4.05 V (R14/R15): FULLCHG_EN -> 4.20 V full-charge mode
LCHG = 1.0 A (R18); RT1 = NCP18XN103 on the holder, 0.45 C window; 4.4 h timer
Charges only on the 15 V contract (UVLO 11.2 V); runs with no cell.

RAILS – 5 V boost (always on) -> 3V3 buck



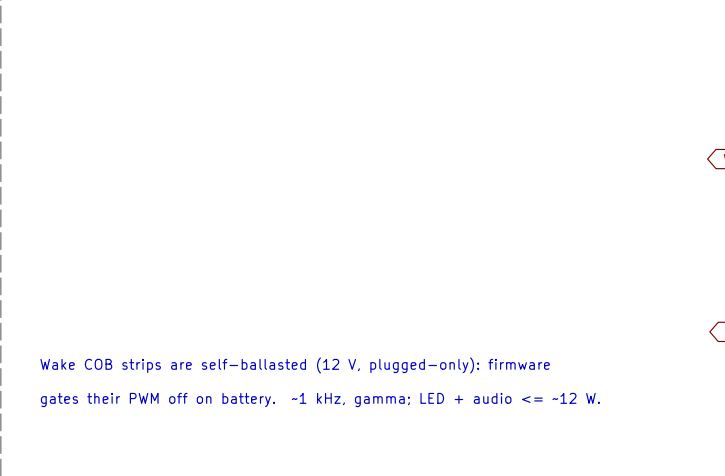
+5V: display panel, stepper VM, panel LEDs, PVDD-mux aux. +3V3: MCU + all logic.

RAIL – 12 V boost (plugged-only: audio PVDD + wake LEDs)



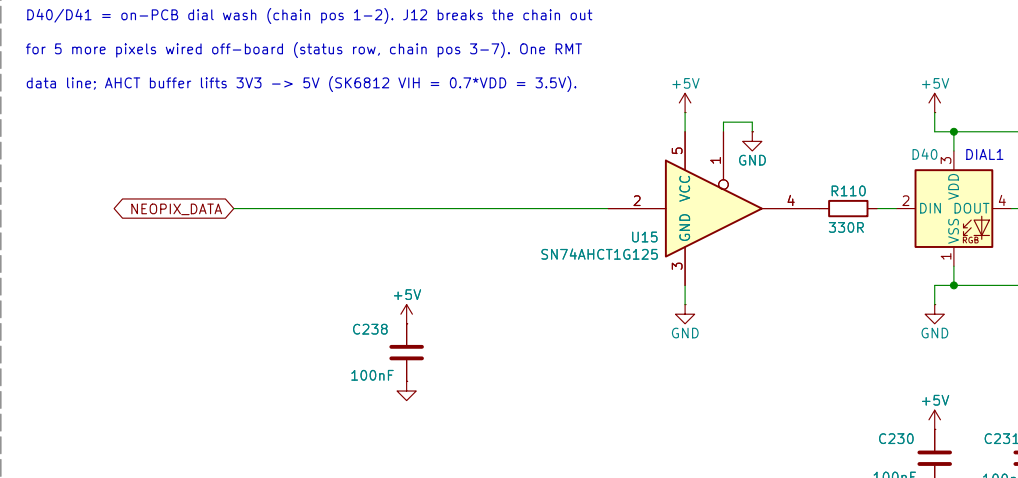
-12 V ceiling from 15 Input: wake LEDs = audio share it.
PLUGGED-ONLY: firmware asserts BOOST12_EN only while PD_PG is live.

LED – wake COB drivers (12 V, plugged-only)



Wake COB strips are self-ballasted (12 V, plugged-only): firmware gates their PWM off on battery. -1 kHz, gamma: LED = audio <= -12 W.

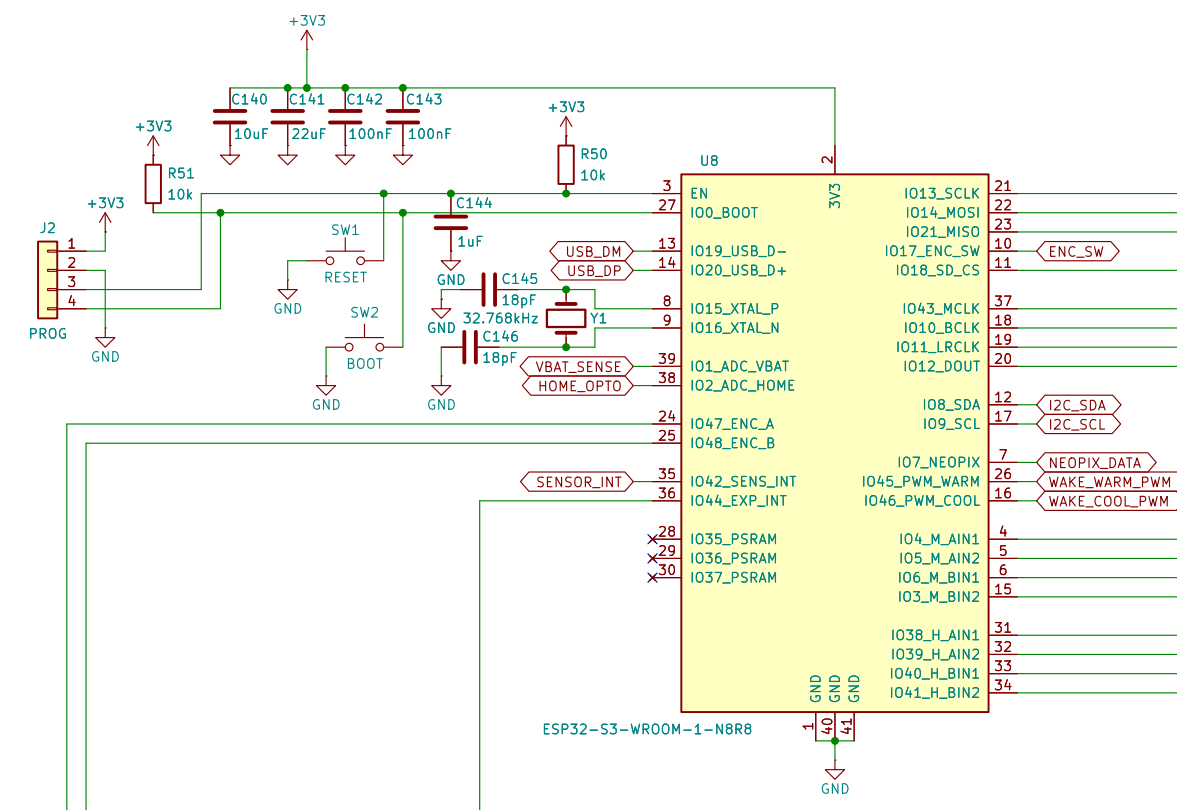
DIAL NEOPIXEL HEAD – 2x on-PCB + 5x off-board via J12 (5 V)



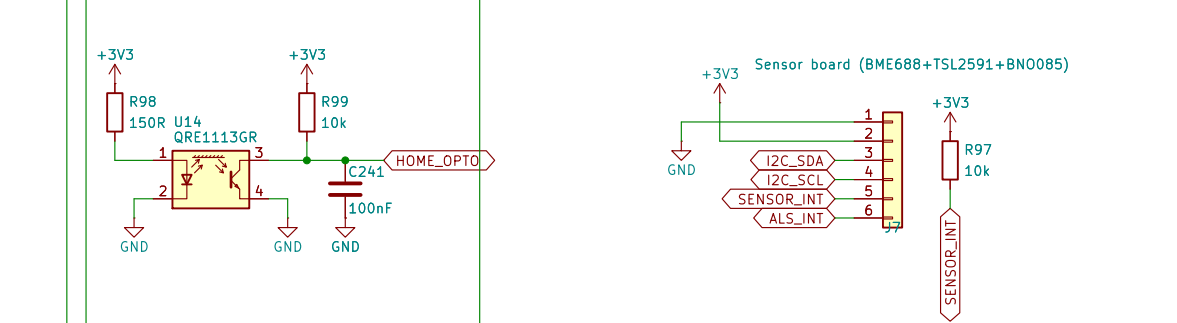
Layout: D40/D41 = dial wash either side of the shaft. C230/231 one at each pixel VDD: C237 bulk + R110 + U15 at the chain head. J12 feeds 5 more SK6812 wired off-board in series (chain pos 3-7), same rail.

MCU – ESP32-S3-WROOM-1-N8R8 (Wi-Fi + BLE, native USB-JTAG)

IO35-37 reserved by the octal PSRAM (N8R8). Straps: IO0 high; IO45/46 low via their wake gate nets + Internal PD; IO3 Internal. Flash+console+JTAG = USB-CDC (IO19/20).

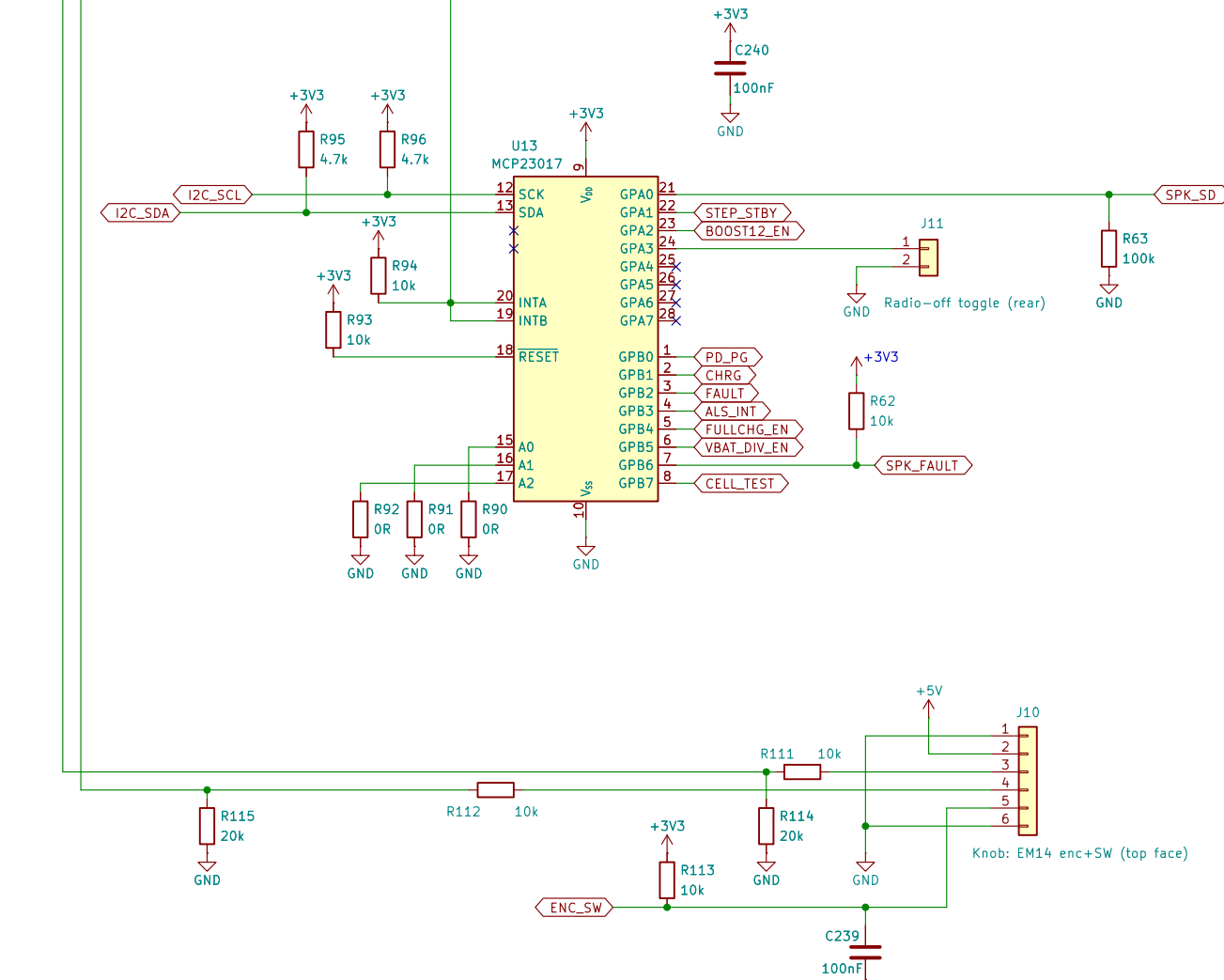


SENSORS (I2C off-board) + HAND-HOMING (QRE1113)



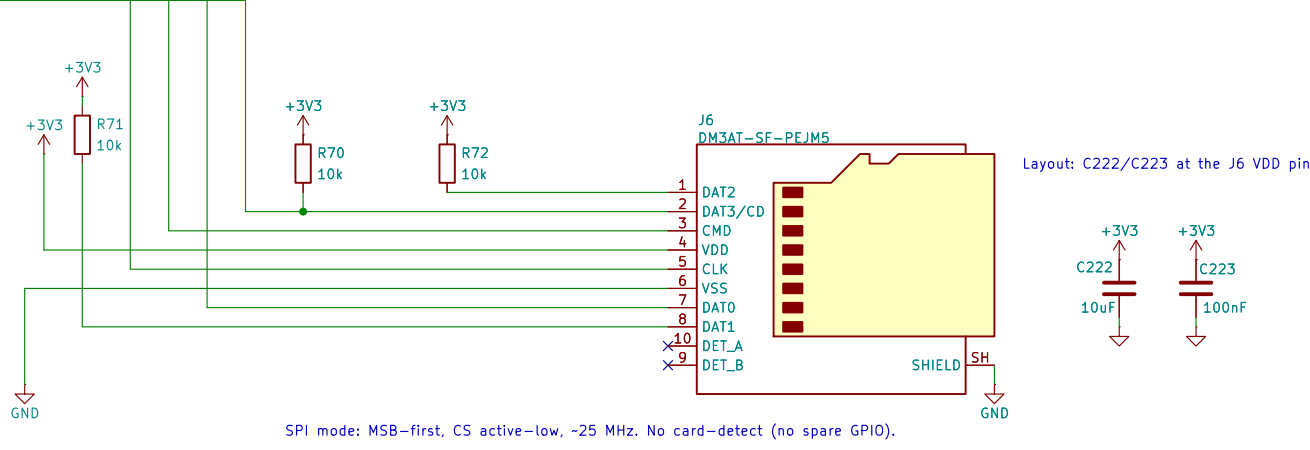
QRE1113GR faces reflective tabs on the hand hubs (no magnets).
Sensor breakouts carry their own I2C pull-ups.

IO EXPANDER (MCP23017 0x20) + KNOB (EM14)



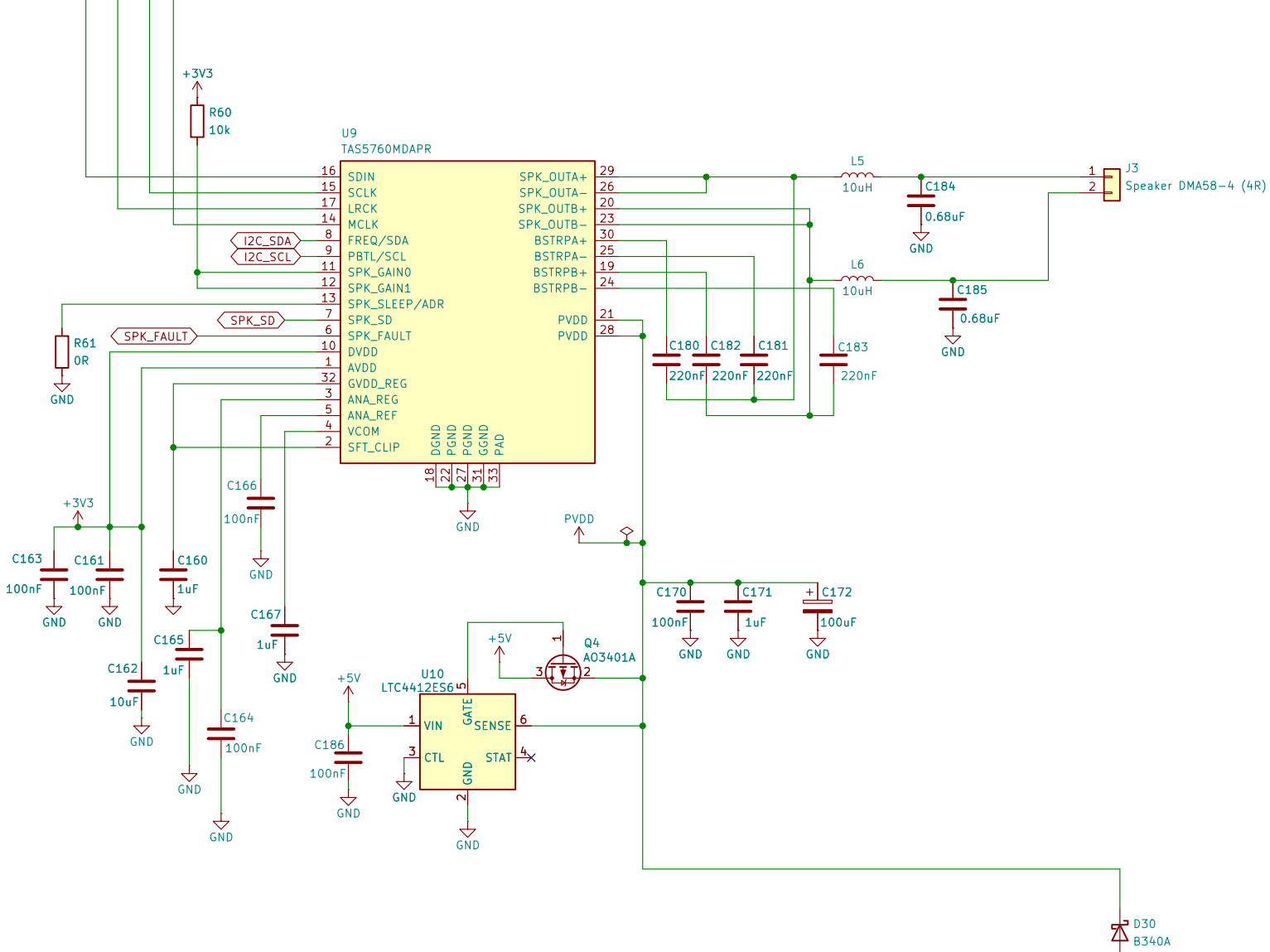
INT: any GPA/GPB change -> IO44 (IOCON.MIRROR=1). Knob = EM14A00=C24-L0645 (optical, 6A CPR, no detent, push) on the top face. J7/J10 = JST 2H 1x06, J11 = 2H 1x02 (pre-crimped Axx2R cables). A/B 5 V -> 100k/200k div -> PCNT; SW -> IO17 IRQ.

STORAGE – microSD (DM3AT), SPI2 dedicated



SPI mode: MSB-first, CS active-low, -25 Mhz. No card-detect (no spare GPIO).

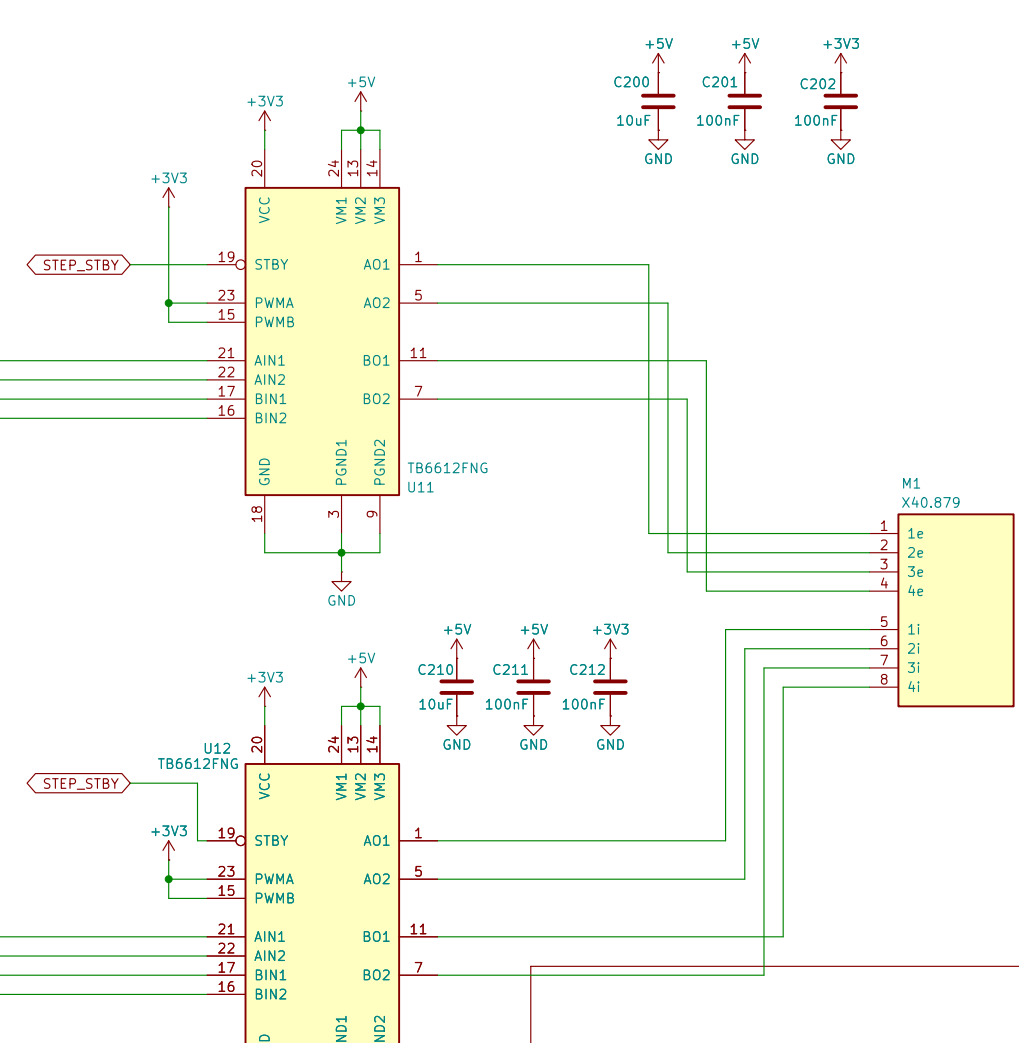
AUDIO – TAS5760M PBTL + LC filter + PVDD mux (12V/5V)



PVDD auto-mux: 12 V wins when plugged; on battery Q4 Ideal-diodes the 5 V rail -> -3 W quieter alarm. Firmware: LED+audio <= -12 W.
PBTL mono (reg 0x06[7]=1): OUTA[1]OUTB. MCLK 256fs -12.288 MHz.

MOTOR – 2x TB6612FNG -> X40.879 dual-shaft stepper

Layout: each driver's 10uF+100nF go at its VM pins, the 100uF at its VCC pin.



PWM-on-IN: microstepping; MCPWM waveforms on AIN/BIN, PWM/A/B tied high.
STEP_STBY (expander GPA1) low at boot -> drivers off. Call map: esp32.md